

EXERCISE 6 –CREATING SHAPEFILES

Create New Shapefiles

1. Add Hambantota.tiff file
2. Zoom your map into Karagan Lewaya
3. Select Create Layer from the Layer menu, and then the New Shapefile Layer tool.
4. This will launch the dialog box below, for creating a new empty shapefile (vector layer).
 - *File name* – this is the name of the shapefile and location where you would like to store it.
 - *File encoding* – this sets the format of the text for specific language characters such as ~ and accent marks. We will use the default value of UTF-8 in our exercise.
 - *Geometry Type (Point, Multipoint, Line, Polygon)* – this determines the type of geometry for the data that you want to capture.
 - *Coordinate System (Specify CRS)* – This allows you to specify the coordinate system that you want to use for your shapefile.
 - *New attribute* – This is where we create the fields (columns) in our attribute table, which allows us to capture data for selected features from the image.
5. Create a new point shapefile using the following settings:
 - Type: Point*
 - New Attribute*
 - Name: Name*
 - Type: Text data*
 - Width: 50*
6. Click the Add to attributes list button to add the new attribute
7. Click OK and save the shapefile to **Buildings.shp**

Your new shapefile layer should be automatically added into the map and appear in the **TABLE OF CONTENTS**.

Creating New Line and Polygon Shapefile Layers

1. Repeat the process used to make a new point shapefile to create a new line shapefile layer with the following settings:
 - Type: Line*
 - New Attribute*
 - Name: Name*

Type: Text data
Width: 50

New Attribute
Name: Length
Type: Decimal number
Width: 13
Precision: 3

2. Click the Add to attributes list button to add the new attribute
3. Click OK and save the shapefile to **Roads.shp**
4. Lastly, create a polygon shapefile with the following settings:

Type: Polygon
New Attribute
Name: Name
Type: Text data
Width: 50

New Attribute
Name: Area
Type: Decimal number
Width: 13
Precision: 3

5. Click the Add to attributes list button to add the new attribute
6. Click OK and save the shapefile to **LULC.shp**
7. We now have our new shapefile layers ready to begin editing.

EXERCISE 7 – CREATING AND EDITING POINTS AND POLYGONS

Editing Toolbar



Creating and Editing Points

1. Select the Buildings layer in the **TABLE OF CONTENTS**
2. Click the Toggle Editing button
3. Select the Add Feature(s) tool 
4. Left click on the map to add a point (it doesn't matter where)
5. Click OK to finish creating the point

We now have our first new point feature in the map. Next, we will move its location.

1. Click the Move Feature(s) tool 
2. Move the cursor over your point
3. Left click and hold over the point
4. Move the cursor to the correct point and release the mouse button

To delete a point, follow these steps:

1. Click the Toggle Editing button (if you are not already in an editing session)
2. Click the Select Single Feature tool  (we used it in an earlier exercise)
3. Click on the point to select it – the point will be highlighted yellow
4. Click the Delete Selected tool  on the Editing toolbar

When you have finished editing the new layer you need to end the editing session and save your edits:

- Click the Toggle Editing button 
- Click Save

Creating and Editing Polygons

1. Select the **Water.shp** layer in the TABLE OF CONTENTS
2. Click the Toggle Editing button
3. Select the Add Polygon Feature(s) tool
4. Left click on the map to add a vertex and then trace the outline of the Water feature by left clicking to add vertices at the four corners of the building.
5. Once you have traced the outline of the building's roof, right click to finish the polygon and add attributes.
6. Click OK to finish creating the polygon
7. We now have polygon features in the map. We can edit, move, and delete polygons using the same tools that we used in our point example above.
8. Please finish road layer editing and add screen shot to the chat box. Please upload your all practical from first day to LMS system.